Assignment 2: Immersive Audio Storytelling

Section 1: Instructions on how to set up and start your patch

1. Keep the whole project folder together (the Max patch, the D4 library files and the sounds). Open Max 9, then open Assignment2.maxpat from the D4 folder.

2. Check Options → Audio Status: correct output device, sampling rate 48000. The patch outputs binaural sound through poly~ D4.dac.cell.binaural.ambeo~ 9 → ezdac ~. For RTV100 the poly~ can be replaced with d4.dac~ 36.

3. Every sound block has a D4.ui.mixer~ channel strip (the vertical fader under the red D4.sound object). Raise all 15 faders to the top — they load at minimum. Adjust the master volume (top of the patch) up until 0 dB so it sounds right.

4. Level check for RTV100: click "test loudest" (top left). It plays the war cry alone, which is the loudest single sound in the piece (97 s). Set the master/system level with this, then click "stop".

5. Click "play". It turns audio on automatically, resets everything and starts the timeline from 0; the number box next to "sec" shows the elapsed time. "stop" fades all sounds out within half a second. Clicking play again restarts from the beginning.

6. Presentation mode (Ctrl/Cmd+Alt+E) shows only play / stop / sec / test loudest / master volume.

If nothing plays: open the Max Console. If it shows "buffer~ • xxx.wav: can't open", the sound files were not found. The 15 sounds are in D4/sounds/edit — either add that folder in Options → File Preferences (or add the D4 folder with "Subfolders" checked). Then close and reopen the patch.

Section 2: Live performance instructions

Null: There is no live performative element. The piece runs fully automated from the play button. The "test loudest" button is only for the level check before playback.

Section 3: Sound files

All files: 48,000 Hz, 16-bit, mono. Listed in order of appearance.
"Timed triggering" = started and ended at fixed times from the play button via delay objects; "radius" = source radius set per sound (1–120°); "ramp-up/down" = custom per-sound ramp values on D4.calc~.

1st | laugh.wav | Children laughing and playing in a village square (0–15 s). | DRY | AUTOMATED MOVEMENT, TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "children laughing and playing outdoors, village square, 15 seconds"

2nd | bird.wav | Small birds chirping and flying past overhead (6–14 s). | DRY | AUTOMATED MOVEMENT, TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "small birds chirping and flying by, close, 8 seconds"

3rd | heaven.wav | Calm, peaceful music bed for the village scene (16–76 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "peaceful heavenly ambient music, soft pads, 60 seconds"

4th | air_loop.wav | Gentle wind, circling around the listener (16–76 s). | LOOPABLE, DRY | AUTOMATED MOVEMENT (360° rotation), TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "soft steady wind blowing, no gusts, loopable, 30 seconds"

5th | bell_dry.wav | Single church-bell strike, close and in front (46 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "single church bell strike, dry, no reverb"

6th | bell_wet.wav | Reverb tail of the same bell, spreading toward the back (46 s). | WET | AUTOMATED MOVEMENT, TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: made in Adobe Audition from bell_dry.wav (reverb, effect-only output)

7th | horn.wav | Distant war horn sounding from behind (76 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "long war horn blast, distant, medieval, 8 seconds"

8th | army.wav | Marching army approaching from behind to the front (77–97 s). | DRY | AUTOMATED MOVEMENT, TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "army marching footsteps and armor, approaching, 20 seconds"

9th | wahhhh.wav | War cry of the charging soldiers, left and right (97 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN, MANUAL TRIGGER (loudest-point test) | Author: ElevenLabs (AI) with prompt "crowd of soldiers shouting a war cry, charging, 15 seconds"

10th | sword.wav | Sword clashes of the battle (101–118 s). | DRY | AUTOMATED MOVEMENT (random positions), TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "metal sword clashing in a battle, several hits, 10 seconds"

11th | pain.wav | Screams of the wounded (103–118 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "man screaming in pain, battlefield, 15 seconds"

12th | blood.wav | Short stab / blood-splatter hits (101–115 s). | DRY | AUTOMATED MOVEMENT (random positions), TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "short stab and blood splatter impact, 3 seconds"

13th | end.wav | Sad ending music after the battle (120–138 s). | DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "sad, slow ending music, strings, 18 seconds"

14th | ashe.wav | Crackling fire among the ashes (paper being crumpled, used as fire) — played from three directions (135–150 s). | LOOPABLE, RECORDED BY ME, DENOISED, DRY | TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: me (recorded with my phone — crumpling paper; denoised and cross-faded for looping in Adobe Audition)

15th | giggle.wav | Women giggle circling quickly around the listener (151–159 s). | DRY | AUTOMATED MOVEMENT (fast rotation), TIMED TRIGGERING, RADIUS, RAMP-UP/DOWN | Author: ElevenLabs (AI) with prompt "single child giggling, creepy, close, 8 seconds"

Techniques summary: automated movement — line-based paths (laugh, bird, bell_wet, army), 360° rotation with metro/counter (air_loop, giggle), random/drunk positions per hit (sword, blood); timed triggering — all 15 sounds; different radius — 1° (battle, music bed) up to 35° (laugh) and 15°→120° (bell_wet); custom ramp-up/down — 2/20 ms (bell_dry) up to 3000/2000 ms (fire); manual trigger — test loudest (war cry); preset + loadmess — all setup values; looping — air_loop.wav and ashe.wav (cross-faded in Audition).

Section 4: Storytelling Concept and Technical Approach

4.1 Story Concept and Inspiration

This project was inspired by *Attack on Titan*, particularly its portrayal of the recurring cycle of peace, violence, and human conflict. My original idea was to tell a story in which an era of violence is followed by peace, only for violence to emerge again, reflecting how history tends to repeat itself. However, I felt that expressing this entire cycle through a short audio experience would either require too much time or make the message difficult to understand.

Instead, I decided to focus on a simpler idea: peace can be broken at any moment. The story begins in a peaceful environment, gradually introduces signs of approaching conflict, and eventually transitions into the chaos of war. The ending reflects on human ignorance, forgetfulness, and greed through a mocking female laugh.

Rather than explaining the story through words, I wanted listeners to experience this transition entirely through sound and interpret its meaning themselves.

4.2 Sound Choices and Narrative Structure

The experience begins with people laughing and birds flying, establishing an environment where people are living peacefully. Gentle wind continues throughout the opening sequence, accompanied by calm, peaceful, almost sacred music. These sounds are intended to create a sense of comfort and safety.

After a brief moment of silence, a war horn interrupts the peaceful atmosphere. The horn is followed by marching footsteps, suggesting that something threatening is approaching. I wanted these sounds to create tension before the actual conflict begins.

The following sequence uses sounds of swords clashing, people suffering, and violent combat to represent the chaos of war. As the battle continues, these sounds gradually fade out, followed by a melancholic piece of music that suggests everything has finally come to an end.

Finally, a woman's laughter is heard. Unlike the laughter at the beginning, this laughter is not meant to express happiness. Instead, it represents a mocking response to humanity's ignorance, tendency to forget the past, and repeated pursuit of conflict driven by greed.

4.3 Spatial Audio and Sound Manipulation

Spatial audio plays an important role in communicating the story because I wanted listeners to feel as though they were physically present within each scene.

At the beginning, I used 360-degree spatial movement for the sounds of people laughing to create the impression of being surrounded by happy people. The bird sounds were spatialized to add depth and make the environment feel more natural. Similarly, the wind was distributed around the listener to create the feeling of standing in the middle of a peaceful outdoor environment.

I also separated the bell sound into dry and wet components to provide more control over its spatial characteristics and reverberation. This was intended to create a greater sense of space and reinforce the peaceful, almost sacred atmosphere.

During the transition to war, the marching footsteps were spatialized to make listeners feel as though they were standing among advancing soldiers. The battle sequence combines multiple combat sounds positioned in different directions, creating the impression of being surrounded by fighting in the middle of enemy territory.

I also used gradual fade-outs during the transition from combat to the final music. Rather than abruptly stopping the battle, the fading sounds were intended to make the end of the conflict feel more natural and emotionally significant.

Finally, the female laughter was spatialized to make listeners feel personally targeted by the mocking laughter, reinforcing the unsettling meaning of the ending.

4.4 Design Rationale

My main goal was to use spatial audio not simply to make the sounds more realistic, but to support the emotional progression of the story.

The opening uses environmental sounds and spatial movement to establish comfort and immersion. As the story progresses, the soundscape becomes increasingly threatening and chaotic. The contrast between these two environments is intended to emphasize how suddenly peace can disappear.

I also wanted the final laughter to change how listeners interpret the experience. While the battle itself appears to have ended, the laughter suggests that the underlying human behaviors responsible for conflict have not necessarily disappeared.

Through these choices, I wanted to communicate that peace is fragile and that humanity may continue repeating the same mistakes, even after experiencing the consequences of violence.

4.5 References

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